

1. Alternations and Decomposition

MAIN QUESTION: Can lexical frequency distribution facilitate learning of a variable phonological alternation?

(1) English Stress Shift: Word prominence shifts rightward when suffixes are added (aka level 1 and level 2 suffixes)

a. párent + al → [parént-al] b. párent + hood → [párent-hood]
Application suffix 😊 Non-application suffix ☹️

Previous analyses claim speakers have internalized application and non-application categories which derive correct phonological patterns [1, 8]

What information do learners use to establish these categories?

Speakers are attuned to differences in **lexical decomposition** and may rely on them to learn idiosyncrasies in morphological productivity [4, 5], phonotactics [6], and phonological alternations [2, 3, 12]

Decomposition: Abstract lexical parsing process part of lexical access; predicted using **token relative frequency** [4, 5, 6]

Decomposed parse vs **Whole-word parse**
/move/ + /-ment/ vs /movement/
Stem freq. > affixed freq. vs Affixed freq. > stem freq.

Hypothesis: Speakers rely on lexical relative frequency distribution to accurately learn phonological processes with varied application

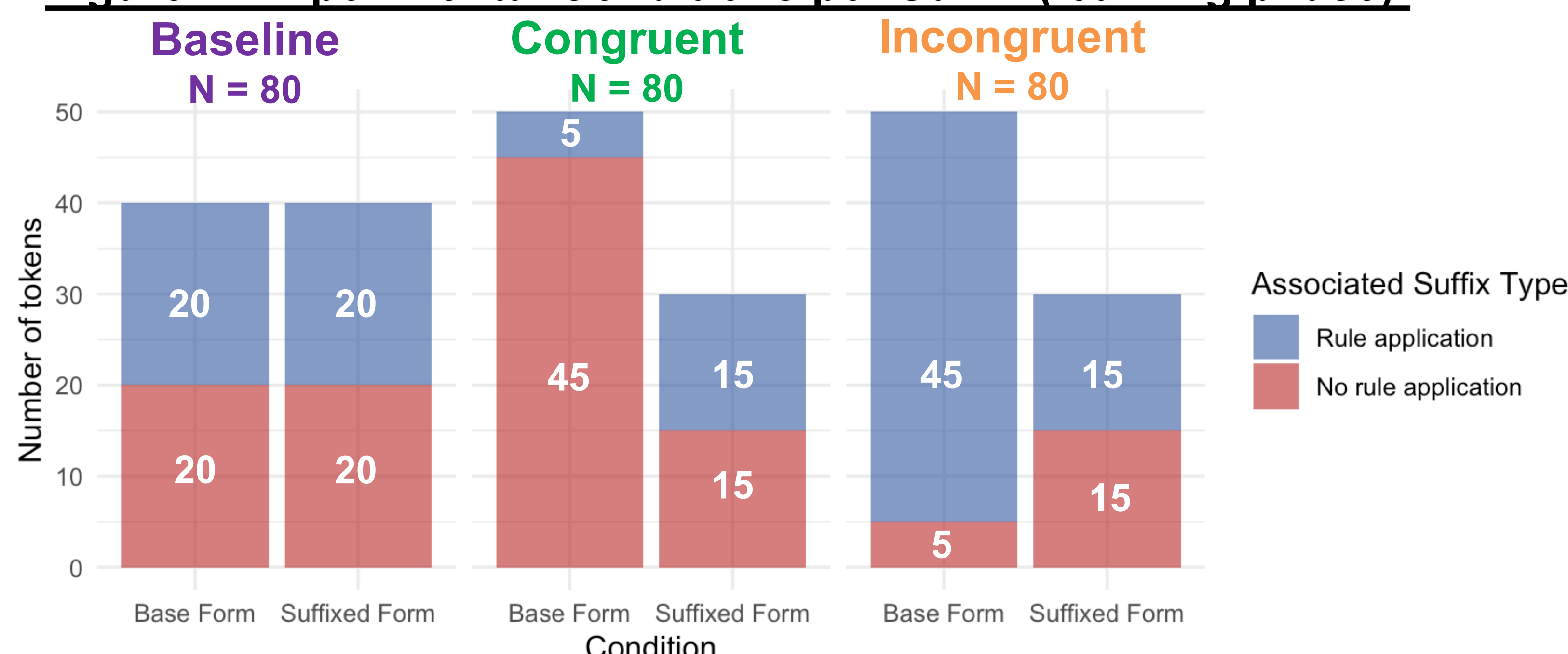
3. Procedure

THIS STUDY: Learning phase shows meanings of stems and suffixes, phonological rule application, and artificial frequency distribution [7, 10]

Stem and suffixed forms presented separately

Stem and suffixed forms presented in **particular relative frequency ratio** which encourages particular parsing strategies for each suffix type

Figure 1: Experimental Conditions per Suffix (learning phase):



Base form freq. > affixed form freq. (decomp.) strengthens *separate* stem and suffix representations (e.g. /move/ + /ment/), making the **non-application pattern** *easier* to learn

Affixed form freq. > base form freq. (whole-word) promotes *combined* stem-suffix representation, making non-application pattern *harder* to learn

Prediction: Test phase response accuracy in **congruent condition** > **incongruent condition**

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2. Participants and Stimuli

Artificial language learning (ALL) study (3 phases)

180 native English-speakers participated in PCIBex [11] experiment, recruited via Prolific [9], excluded 13 → **167 total participants**

Nonce word stems (CVCVC) and 2 nonce suffixes (VCV) presented via audio and orthography

Stem-final voicing assimilation rule applies with one suffix but not the other
[porap] + [-iso] → [porab-iso] (**assimilation applies – application suffix**)
[porap] + [-ifa] → [porap-ifa] (**no assimilation – non-application suffix**)

All stimuli associated with meanings:

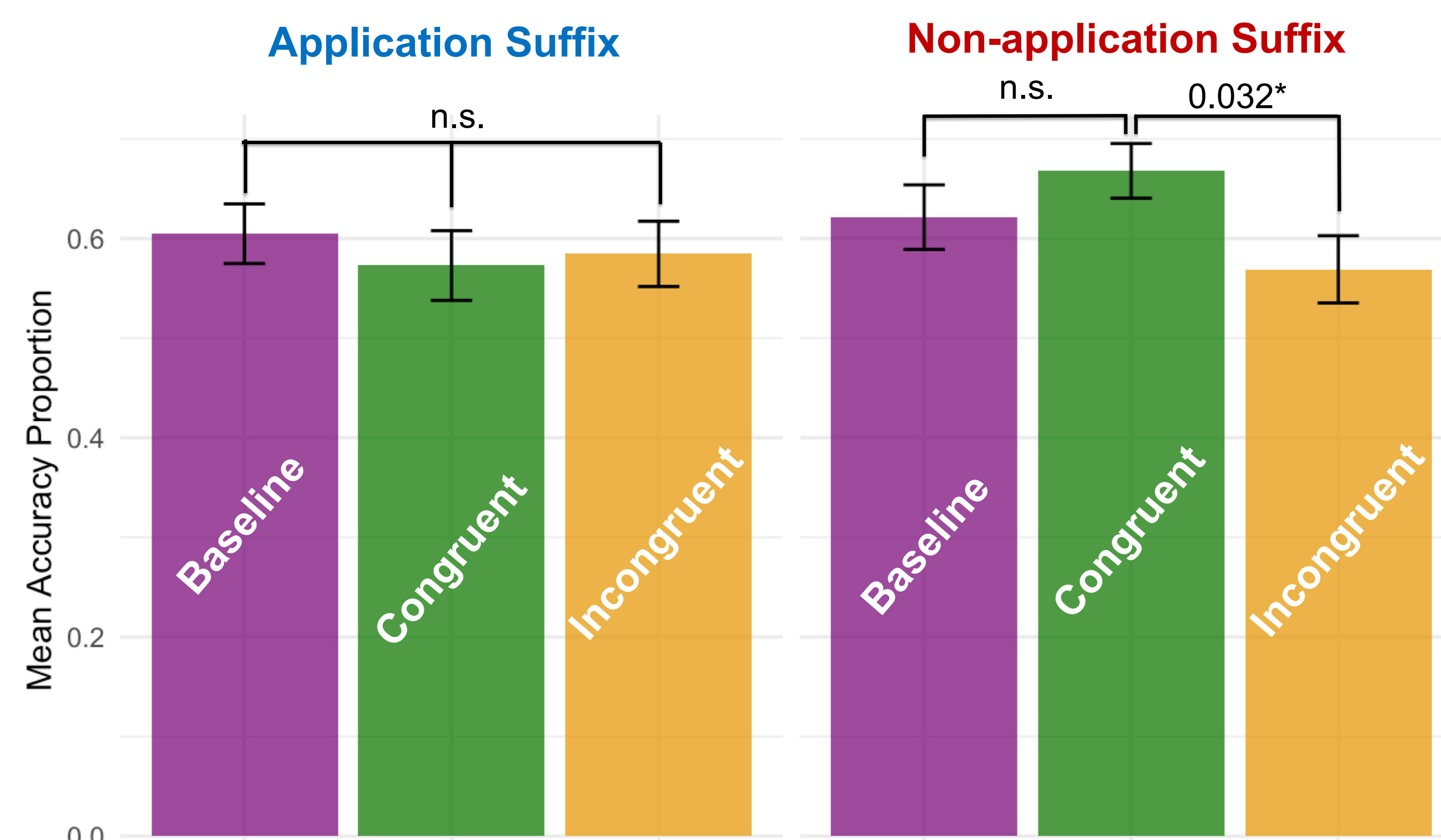
Stem = objects and animals, suffix 1 = 'small', suffix 2 = 'big'

Suffix meanings and application/non-application counterbalanced

138 total trials (88 learning, 20 verification, 30 test)

4. Results

Figure 2: Accuracy for Each Suffix Type Between Conditions:



For **application-suffixed items**, generalized linear mixed effects model found no statistically significant accuracy differences between conditions

For **non-application-suffixed items**, accuracy was significantly higher in **congruent** condition than in **incongruent** condition ($\hat{\beta} = -0.50$ $p = 0.032$)

5. Discussion and Conclusions

- Results showed learning of non-application pattern was facilitated in the **congruent** condition compared to the **incongruent** condition
 - To reach higher accuracy levels in learning, speakers need sufficient exposure to base forms for non-alternating suffixed forms when an alternating form is also generalizable
 - Same exposure is not needed for alternating suffixed forms
- Aligns with distributional cues to lexical decomposition – suggests **lexical parsing processes like decomposition help speakers learn which forms should/should not undergo alternations**
- Supports phonological theories which incorporate distributional lexical information into phonological representations

Remaining Q: Why wasn't application-suffix accuracy improved by suffixed form > base form distribution?

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